Musanze Harvest Intelligence Service: Technical Report

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1. Identity and decision problem

Item	Value
Full name	Manzi Kassim
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Verification code	AI-A2-7935
Seed (LLLL)	7935
Repository	https://github.com/manzikassim/AI_A2_2527935
Final commit hash	COMMIT_HASH_HERE

Stakeholder. A potato cooperative in Musanze whose field officer receives farm records in inconsistent spreadsheets and needs an evidence-based tool that works offline.

Prediction questions.

1. How many tons of potatoes will a farm harvest? (regression)
2. Is a farm at high production risk, meaning its yield per hectare is below the median? (classification)
3. Which farms operate under similar conditions? (clustering)

Ethical boundary. This is a classroom decision-support prototype trained on synthetic data. It does not replace an agronomist and must not be used to decide access to finance, insurance or public services. A human must review every output.

2. Data and reproducibility

The dataset is synthetic and is recreated by `data/generate_data.py` from the personal seed 7935 using `numpy.random.default_rng(7935)`.

Column	Rule	Role
rainfall_mm	Uniform 600 to 1400	Feature
avg_temp_c	Uniform 12 to 26	Feature

Column	Rule	Role
soil_moisture_pct	Uniform 20 to 80	Feature
fertilizer_kg	Uniform 0 to 250	Feature
field_size_ha	Uniform 0.2 to 4.0	Feature
season	Random A or B	Feature
cooperative_support	Random 0 or 1	Feature
yield_tons	Given formula, noise SD 0.80, minimum 0.50	Regression target
high_risk	Yield per hectare below the median	Classification target

Row count. Base rows = 240 + (7935 mod 41) = 240 + 22 = 262. After appending 3 exact duplicates the raw file has 265 rows.

Deliberate data problems. With m = round(0.05 × 262) = 13, the generator set 13 values missing in soil_moisture_pct and 13 in fertilizer_kg. The validation report found exactly these 26 missing values and 3 duplicate rows, with no out-of-range values. The pipeline drops the duplicates (265 to 262 rows) and imputes missing values with the training median inside the model pipeline.

Fingerprint. SHA-256 of the final CSV (written with four decimals and `\n` line endings so it is identical on Windows and Linux):

`6535cb925d58e2a4ca2bee57af978029da6538103735682eb5d26fc014b58863`

Running `run_all.py` twice reproduced the same fingerprint and identical metric files.

3. Pipeline design

- **Validation first:** `run_all.py` rejects a seed that is not the last four digits of the student ID, then checks required columns, ranges, missing values, duplicates and leakage.
- **Fixed split:** 262 clean rows split 209 train and 53 test with `random_state=7935`, stratified on high_risk. The same split serves all three tasks so results are comparable.
- **Preprocessing (train-only):** a ColumnTransformer applies median imputation and standard scaling to the five numeric features, one-hot encoding to season, and passes cooperative_support through. It is fitted on training data only and is part of every saved Pipeline, so inference applies exactly the same steps.
- **Leakage controls:** yield_tons, yield per hectare and high_risk can never be input features (`check_leakage`). Duplicates are removed before the split so identical rows cannot sit in both train and test.
- **Candidates:** regression uses a mean baseline, linear regression and a random forest. Classification uses a majority baseline, logistic regression and a random forest. Clustering uses K-means.

- **Selection rule:** candidates are compared by 5-fold cross-validation on the training set only (RMSE for regression, F1 for classification). The test set is used only for the final report, so it cannot influence the choice.
- **Random states:** all set to 7935.

4. Regression and classification

4.1 Regression results (test set, 53 rows)

Model	MAE	RMSE	R^2	CV RMSE (train)
Mean baseline	2.0050	2.5538	-0.0043	2.5709
Linear regression (selected)	**0.7520**	**0.9271**	**0.8676**	**1.0761**
Random forest	0.9145	1.1801	0.7855	1.4166

Linear regression was selected because it had the lowest cross-validated RMSE (1.0761) on the training set. It also had the best test scores. Its RMSE of 0.93 tons is close to the noise standard deviation of 0.80 built into the data, so little improvement is possible.

Residual plot: insert `artifacts/plots/residuals.png` here (Figure 1).

One important error. Test row 107 had the largest error. Actual yield 11.4745 tons, predicted 13.8444 tons, error +2.3699 tons (over-prediction). Its conditions: rainfall 1253.5 mm, temperature 13.73 °C, soil moisture 51.6 %, fertilizer 22.0 kg, field size 3.29 ha, season A, with cooperative support.

Interpretation: Row 107 was over-predicted by 2.3699 tons, which is almost three times the data's noise standard deviation of 0.80 tons, so this is a relatively large error. The farm had low fertilizer use, a cool temperature and a fairly large field, and the model may not have represented the combined effect of these conditions accurately. In particular, a linear regression model represents temperature with a straight-line effect, so it may not capture a curved or optimal-temperature relationship in the yield formula. This example shows that even a model with good overall scores can make substantial errors for individual farms.

4.2 Classification results (test set, 53 rows)

Model	Precision	Recall	F1	ROC-AUC	CV F1 (train)
Majority baseline	0.4906	1.0000	0.6582	0.5000	0.5333
Logistic regression (selected)	**0.9259**	**0.9615**	**0.9434**	**0.9915**	**0.9280**
Random forest	0.9286	1.0000	0.9630	0.9986	0.9204

Confusion matrix of the selected model (rows = actual, columns = predicted): low risk 25 correct and 2 wrongly flagged; high risk 1 missed and 25 correct.

Model choice. The random forest scored slightly higher on the test set (F1 0.9630), but logistic regression had the higher cross-validated F1 on the training data (0.9280 against 0.9204), so it was selected. Choosing by test score would have used the test set to pick the model and made the test result optimistic. The majority baseline reaches recall 1.0 only by flagging every farm as high risk, which is why its precision and ROC-AUC are poor.

Confusion matrix plot: insert `artifacts/plots/confusion_matrix.png` here (Figure 2).

One important error. The selected model made three mistakes:

Row	Actual	Predicted	Risk probability	Yield per ha
37	not high risk	high risk	0.5412	6.4648
223	not high risk	high risk	0.6710	5.8835
0	high risk	not high risk	0.4905	5.3704

Interpretation: Row 37 was flagged as high risk even though its actual label was not high risk; its yield was 6.4648 tons per hectare and the model assigned a risk probability of 0.5412. The probability is close to 0.5, meaning the prediction was uncertain rather than a strong indication of risk. Farms near the median cutoff can be difficult to classify because small changes in yield or field size can move the yield-per-hectare ratio across the cutoff, while random noise also affects yield. A field officer should therefore review such borderline cases instead of treating the predicted label as a definite conclusion.

5. Clustering

Features were imputed, scaled and one-hot encoded with the training-fitted preprocessor, and K-means was run for K = 2 to 6 on the training data without using any target.

K	2	3	4	5	6
Silhouette	0.1437	0.1303	0.1446	0.1422	**0.1479**

K = 6 has the highest silhouette score, but only by a very small margin, and all scores are low (below 0.25, a common threshold for weak structure). This is expected, because the features were generated independently and uniformly, so the data contain no strong natural groups. The clusters are best read as practical groupings by dominant conditions, not as distinct farm types.

Cluster plot: insert `artifacts/plots/clusters.png` here (Figure 3).

Cluster	Farms (train)	Profile (differences from the training average, in standard deviations)	Mean yield (post hoc)
0	36	Much lower rainfall (-1.14), warmer (+0.84), larger fields (+0.51)	11.96
1	28	Low soil moisture (-0.97), high fertilizer (+0.96), larger fields (+0.74), more rain (+0.53)	14.06
2	40	Warmer (+0.90), much smaller fields (-1.03)	9.98

Cluster	Farms (train)	Profile (differences from the training average, in standard deviations)	Mean yield (post hoc)
3	34	Wetter soil (+0.85), more rain (+0.75), low fertilizer (-1.13), cooler (-0.65)	12.62
4	28	Drier and cooler overall: rainfall -0.92, soil moisture -0.75, fertilizer -0.74, temperature -0.70	10.04
5	43	High fertilizer (+0.95), moist soil (+0.53), cooler (-0.75), smaller fields (-0.46)	13.42

Mean yield is shown only to help interpretation. It was not used to form the clusters.

Interpretation: Cluster 1 represents farms that tend to have relatively low soil moisture, higher fertilizer use, larger fields and somewhat higher rainfall than the training-set average. A field officer could check whether the fertilizer is being applied efficiently and assess whether irrigation or soil-moisture conservation would help these farms. The cluster's mean yield of 14.06 tons is descriptive only; yield was not used to create the cluster. The clusters overlap and their silhouette scores are low, so this group should be treated as a practical comparison group rather than a distinct or definitive farm type.

6. API and testing

Endpoint contract.

Endpoint	Purpose	Success	Failure
GET /health	Liveness and model-loaded check, returns verification code	200	none expected
GET /model-info	Student ID, seed, fingerprint, selected models, metrics, cluster profiles	200	503 if models are missing
POST /predict	Validated farm record in, yield, risk, probability, cluster and verification code out	200	422 for invalid input, 503 if models are missing

The request uses a typed Pydantic model with range limits, restricts season to A or B and cooperative_support to 0 or 1, and rejects unknown fields.

Valid request (`samples/valid.json`) returned status 200 with yield_estimate_tons 13.3541, high_risk 0, risk_probability 0.0335, cluster 5 and verification_code AI-A2-7935.

Invalid request with rainfall_mm 5000 returned status 422 with the message that the input should be less than or equal to 1400. The command-line `predict.py` gives the equivalent clear error and exit code 2.

Tests. `pytest -q` ran 16 tests on Windows 11 Pro 22H2 with Python 3.14.3 and all passed (about 16 seconds, one library deprecation warning). They cover: valid inference, a missing field, a wrong type, a range error, a bad category, dataset reproducibility (same SHA-256 twice, 265 rows, 13 + 13 missing, 3 duplicates), metric reproducibility, rejection of a wrong seed, the leakage guard, no train/test overlap, and six API tests (health, model-info, valid predict, and 422 responses for a range error, a wrong type and a missing field). Most tests exercise failure paths.

7. Model card and reflection

Intended use. A classroom demonstration of yield estimation, risk flagging and farm grouping for a potato cooperative, to support a field officer's discussion, not to make decisions.

Data status. Entirely synthetic, from a fixed formula plus noise. It contains no real farms.

Limitations.

- Only five conditions plus season and support are modelled; real yields depend on soil type, variety, pests, disease and management.
- Results on synthetic data say nothing about accuracy on real farms.
- Clusters are weak (silhouette about 0.14).
- The test set has only 53 farms, so metrics have wide uncertainty.
- Inputs are limited to the training ranges, and the API rejects values outside them.

Risks. Over-trusting a confident-looking number, treating high_risk as a judgement about a farmer, and applying the model outside its training ranges or to different crops or regions.

Prohibited use. Decisions about loans, insurance, subsidies or any public service, and any use without human review.

Human oversight. An agronomist or field officer must review every prediction, and every response carries a disclaimer and the verification code.

What I would improve with real data. Collect real farm records with consistent definitions, validate against measured harvests, check fairness across areas and farm sizes, add uncertainty intervals, and test whether a threshold other than the median suits the cooperative's actual risk definition.

Reflection: I learned that model selection should be based on cross-validation on the training data, while the test set should be kept for the final evaluation to avoid optimistic results. I was also surprised that a model with good overall metrics could still make a large error for one farm, and that clustering can produce weak groups when the input features do not naturally form distinct patterns. The hardest part was interpreting these results responsibly and explaining why predictions need to be reviewed by a field officer.